Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH (ECE)-V
Name of the Paper: Data Communication Network
Paper Code: TEC-505
Time: 1.5 hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1.

(10 Marks) CO1

a. Discuss the fundamental goals of data communication and explain how these goals apply to real-world network applications.

OR

b. Explain the concept of wireless networks. Discuss their advantages and disadvantages compared to wired networks, and describe the common applications of wireless technology in today's society.

Q2.

(10 Marks) CO1

a. Discuss the TCP/IP model and its importance in modern networking. Compare and contrast it with the OSI model, focusing on the layers and protocols involved in data communication.

OR

b. Define data and signals in the context of data communication. Explain the differences between analog and digital signals, and discuss the advantages and disadvantages of each type.

Q3.

(10 Marks)CO1,2

a.Define bandwidth and discuss the significance of bandwidth utilization in communication networks. Explain the techniques of multiplexing and spreading, providing examples of each and their applications.

OR

b.Discuss the importance of error control in data link protocols. Explain the methods used for error detection and correction, and illustrate your answer with examples of common techniques.

Q4.

(10 Marks) CO2

a. Calculate the CRC for the data word 10110110 and divisor be 1011 which is agreed upon by the sender and the receiver. Show the steps performed at sender and the destination end.

OR

b. Describe elementary data link protocols, including the Unrestricted Simplex Protocol and the Simplex Stop-and-Wait Protocol. Discuss their functionalities and limitations in a networking context.

Q5.

(10 Marks) CO2

a. Explain HDLC (High-Level Data Link Control) in detail with the help of a diagram.

OR

b. Explain the concept of check summing method with the help of an example.